EXPERIMENT-9

9. Scenario: You work for a real estate agency and have been given a dataset containing information about properties for sale. The dataset is stored in a Pandas DataFrame name property_data. The DataFrame has columns for property ID, location, number of bedrooms, area in square feet, and listing price. Your task is to analyze the data and answer specific questions about the properties.

Question: Using Pandas DataFrame operations, how would you find the following information

from the property_data DataFrame:

1. The average listing price of properties in each location.

2. The number of properties with more than four bedrooms.

3. The property with the largest area.

Aim

To analyze property data using Pandas and determine the average listing price for each location, the number of properties with more than four bedrooms, and the property with the largest area.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame containing property IDs, locations, bedrooms, areas, and listing prices.
3. Group the properties by location and calculate the average listing price.
4. Filter the properties having more than four bedrooms and count them.
5. Find the row with the maximum area using idxmax().
6. Display the results.

Code

```
import pandas as pd
```

```
property_data = pd.DataFrame({
    'Property_ID': [101, 102, 103, 104, 105],
    'Location': [
        'Chennai', 'Bangalore', 'Chennai',
        'Hyderabad', 'Bangalore'
    ],
    'Bedrooms': [3, 5, 4, 6, 5],
    'Area_sqft': [1200, 2000, 1800, 2500, 2200],
    'Listing_Price': [2500000, 4500000, 3500000,
                      5500000, 5000000]
})
# 1. Average listing price by location
average_price = property_data.groupby(
    'Location'
)['Listing_Price'].mean()
# 2. Count properties with more than four bedrooms
count_properties = (
    property_data['Bedrooms'] > 4
).sum()
# 3. Property with the largest area
largest_property = property_data.loc[
```

```
    property_data['Area_sqft'].idxmax()
]
print("Average listing price by location:")
print(average_price)
print("\nProperties with more than 4 bedrooms:",
      count_properties)
print("\nProperty with the largest area:")
print(largest_property)
```

Input

```
Property_ID  Location   Bedrooms  Area_sqft  Listing_Price
101          Chennai    3         1200       2500000
102          Bangalore  5         2000       4500000
103          Chennai    4         1800       3500000
104          Hyderabad  6         2500       5500000
105          Bangalore  5         2200       5000000
```

Output

```
Average listing price by location:
Location
Bangalore    4750000.0
Chennai      3000000.0
Hyderabad    5500000.0
```

Properties with more than 4 bedrooms: 3

Property with the largest area:

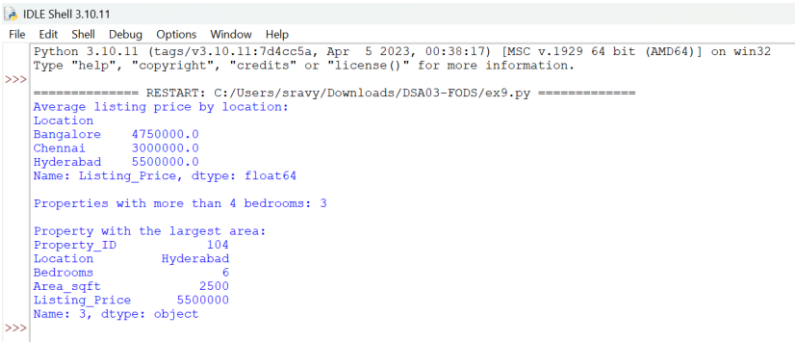
Property_ID 104

Location Hyderabad

Bedrooms 6

Area_sqft 2500

Listing_Price 5500000



Result

Thus, the average listing price for each location, the number of properties with more than four bedrooms, and the property with the largest area were determined successfully using Pandas.